

Ian Gault, MDS MSc RPBio

Applied Data Scientist • Analytical Chemistry & Toxicology

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PROFESSIONAL SUMMARY

Master of Data Science and Registered Professional Biologist with an analytical chemistry research background and six years of experience translating complex chemical data into decision-ready evidence as a consultant. MSc research characterized unknown chemical mixtures using high-resolution mass spectrometry, building experience with high-dimensional instrument data, measurement variability, and censored data. Combines measurement-science foundation with Python/PyTorch ML, computer vision, AWS model training and deployment, and reproducible workflows using Docker, Git, and CI/CD.

EDUCATION & LICENSURE

Master of Data Science (MDS)

SEP 2025 – JUN 2026

University of British Columbia, Vancouver, BC

- **Project Work:** Built CNNs in PyTorch and applied pretrained ResNet50/text embeddings to image data; trained and evaluated classification models; processed datasets locally with DuckDB/Arrow before scaling to distributed PySpark on AWS; and deployed containerized ML workflows using EC2, Docker, automated testing, and CI/CD.

Certificate - Applied Data Science: Leveraging AI for Decision-making

SEP 2023 – DEC 2023

Massachusetts Institute of Technology, Cambridge, MA

Designation - Registered Professional Biologist (RPBio)

2024 – PRESENT

College of Applied Biologists (CAB), BC, Canada

MSc - Analytical & Environmental Toxicology; BSc - Biological Sciences (with Distinction)

2009 – 2019

University of Alberta, Edmonton, AB

PROFESSIONAL EXPERIENCE

Environmental Scientist

JUL 2022 – AUG 2025

WSP, Vancouver, BC

- Delivered environmental monitoring studies and regulatory reporting for 10+ clients across metals, coal, and wastewater sectors, managing multiple concurrent programs as lead data analyst across the full project lifecycle — proposal, execution, analysis, and reporting — contributing to contract renewals exceeding \$200k.
- Initiated the team's shift from manual Excel-based analysis to reproducible, Git-versioned collaborative development and mentored junior staff; replaced ad hoc QA/QC review with a rule-based pipeline for multiple projects.
- Built reusable R frameworks (ETL, harmonization, statistical analysis, reporting) adopted across 2–3 projects, and automated conversion of analyses through to interactive HTML reports and client-ready tables, for senior review.
- Applied statistical and machine-learning methods — trend detection, spatial/multivariate analysis, supervised and unsupervised learning — to flag environmental changes and support regulatory determinations.
- Served as lead data scientist on a small team, advising senior leadership on study design and analytical approach.

Environmental Specialist

MAY 2019 – JUN 2022

Hatfield Consultants, North Vancouver, BC

- Applied a weight-of-evidence approach across contaminated-site ecological risk assessments and monitoring projects, synthesizing complex datasets from water, sediment, and biota into defensible reports for regulators.
- Coordinated end-to-end data pipelines — from field sample collection and contract lab analysis to R-based ETL and analysis-ready outputs — for 10+ clients, gaining familiarity with provincial and federal regulatory requirements.

- Contributed to study design and field data collection across 5+ projects, and stepped into an expanded assistant project manager role across 3 concurrent programs during a period of team understaffing, meeting deadlines.

Graduate Research Assistant

JAN 2015 – JAN 2019

University of Alberta, Edmonton, AB

- Characterized complex, unknown chemical mixtures using high-resolution Orbitrap mass spectrometry, linking analytical signals with biological responses through effect-directed analysis and toxicity identification evaluation.
- Developed and evaluated experimental workflows integrating chemical measurements with in-vitro toxicity assays, working with high-dimensional instrumental data and the measurement variability inherent to laboratory systems

SELECTED DATA SCIENCE PROJECTS

Applied Statistical Modelling Portfolio: Environmental Health & Toxicology

JUN – AUG 2026

Independent project @ iangault.github.io/vignette-non-linear-trends/

- Built a 9-part reproducible R/Quarto portfolio translating consulting-industry water chemistry workflows into a documented data science practice: regression framework, nonlinear trend detection (GAM, GAMM, changepoint, piecewise, LOESS/K-NN) with AIC-based model selection and residual diagnostics (heteroscedasticity, normality, autocorrelation), PCA-based multivariate screening, regulatory compliance screening, and data harmonization.

Multi-source Clinical Data Pipeline & Dashboard

MAY – JUN 2026

BC Children's Hospital Research Institute (UBC MDS Capstone), Vancouver, BC

- Designed a reproducible, modular analysis pipeline in R and a complex, multi-tab interactive Shiny dashboard to characterize Crohn's disease using microbiome, patient, and dietary data, supporting funding and recruitment.
- Built ETL workflows for heterogeneous sources, delivering ML-ready structured datasets and decision-ready outputs via a Makefile-driven environment; authored a companion Quarto blog documenting the statistical analysis pipelines.

Selected Environmental Health & Statistical Modelling Consulting Projects

2020 – 2025

WSP & Hatfield Consultants, Vancouver, BC

- Modelled 20+ years of water-quality chemistry data using Mann-Kendall, GAMs, ANOVA, PCA, and regression to distinguish temporal and spatial patterns associated with wastewater, salinity, flow conditions, and other drivers.
- Developed a food-chain bioaccumulation model integrating soil, water, and dietary chemistry across ~160 spatial plots and multiple wildlife receptors; outputs supported ecological risk assessment and post-closure remediation.
- Analyzed complex observational datasets with detection limits, missingness, outliers, non-normal distributions, and heterogeneous sampling designs, applying assumption testing and parametric and non-parametric methods.

TECHNICAL SKILLS

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| • Programming: R, Python, SQL, Bash | • Data Engineering: databases, ETL, cleaning, feature engineering, data quality | • Machine Learning: Classification, clustering, PCA, regularization, neural networks, deep learning (PyTorch, scikit-learn), agentic coding |
| • Cloud compute: AWS | | |
| • Spatial & temporal analysis: vectors, modelling, time series | • Stats: Descriptive & inferential stats, regression (LMs/GLMs), survival, experimental design | • Visualization & Communication: Microsoft suite, Shiny, heatmaps, Quarto, scientific writing, presentations, decision support |

ACADEMIC PUBLICATIONS & SELECTED RECOGNITIONS

Gault IGM, Sun C, Martin JW. Persistent Cytotoxicity and Endocrine Activity in the First Oil Sands End-Pit Lake. ACS EST Water 2023, 3(2), 366–376 • Morandi GD et al. (incl. Gault IGM). Effects-Directed Analysis of Dissolved Organic Compounds in OSPW. Environ. Sci. Technol. 2015, 49(20) • 3× conference presentation awards (2018): 1st & 3rd place oral presentations at SETAC North America and Prairie chapters • Upper Bound Conference, Amii — Alberta Machine Intelligence Institute, 2024.